

Chapter 11

The Dual Role of Artificial Intelligence in Online Disinformation: A Critical Analysis



Abstract This chapter critically analyses the multifaceted role of artificial intelligence (AI) in the realm of online disinformation. Specifically, it delves into the potential of AI techniques to create highly realistic disinformation and efficiently disseminate it across a vast audience on social media platforms (SMPs). Moreover, the study explores the use of AI as a means to combat this deleterious phenomenon, while simultaneously examining the associated challenges and ethical quandaries. To address these complex issues, the chapter proposes a comprehensive set of recommendations aimed at mitigating these concerns. The chapter concludes that ethical considerations must be an integral part of the development and implementation of AI tools, given the inadvertent negative consequences that might arise from their use. This includes addressing the inherent limitations that give rise to ethical concerns and counterproductive outcomes. The chapter contributes to the academic discourse by addressing the multifaceted challenges posed by AI-generated disinformation.

Keywords Artificial intelligence · Machine learning · Online disinformation · AI · ML · Deepfakes · Recommendation algorithms · Social bots

11.1 Introduction

False, inaccurate, and misleading information deliberately designed, presented, and promoted to manipulate individuals and undermine democratic processes constitutes a longstanding phenomenon [3, 10]. However, in recent decades, the pervasiveness of disinformation has surged significantly, coinciding with the emergence of novel technologies, notably AI [2, 3, 11, 16, 21, 23, 28, 43]. AI can be defined as the capacity of a computer system to perform tasks akin to human intelligence, encompassing learning and decision-making abilities [20, 41]. Machine Learning (ML), a sub-field of AI, entails training computer systems on extensive datasets to recognise undefined patterns and discern valuable information without explicit programming [23, 41, 44]. This AI capability has become highly appealing to malicious actors as it empowers them to fabricate and disseminate highly realistic disinformation, which

rapidly reaches a vast audience [2, 3, 16, 21, 24, 43]. Conversely, AI technologies have also been recognised as potent tools to counter disinformation, enhancing the efficiency and accuracy of detection and removal approaches [3, 21, 28].

This chapter critically examines the dual nature of AI, emphasising its role in both propagating and countering online disinformation, while also examining the ethical implications associated with its application in this context. To this end, the chapter explores the mechanisms of deep learning (DL), with a particular focus on the Generative Adversarial Network (GAN) framework, which plays a pivotal role in creating authentic yet deceptive content. While existing research often focuses on visual and audio deepfakes, this chapter accentuates the significant potential of AI, exemplified by GPT-2, in generating textual content that simulates genuine news articles. This capability effectively blurs the distinction between human- and AI-generated content [16, 23, 24, 44]. The primary contribution of this chapter lies in illuminating the manner in which AI-generated disinformation can induce confusion and foster mistrust in users by convincingly mimicking human-generated texts [16, 23, 24]. Furthermore, it explores the convergence of diverse content formats, where the amalgamation of media enhances the perceived credibility of synthesised information [2, 37]. By elucidating these complex dynamics, the chapter provides valuable insights into the evolving strategies of disinformation campaigns and underscores the urgency of addressing the associated challenges in an increasingly AI-driven information landscape.

The subsequent sections of this chapter are structured as follows: Sect. 11.2 delves into the role of AI in the generation of disinformation, while Sect. 11.3 examines the role of AI in its dissemination. In Sect. 11.4, the role of AI in countering disinformation is examined, with a critical analysis of the associated challenges. Moving forward, Sect. 11.5 presents recommendations that can be implemented to mitigate these challenges and address the ethical concerns surrounding the use of AI. Finally, Sect. 11.6 presents a discussion and conclusion.

11.2 The Role of AI in the Production of Disinformation

In light of the rapid advancements within the domain of AI, a burgeoning concern revolves around the use of these techniques for the automatic generation of disinformation through the manipulation and autogeneration of fake content across various formats [1–3, 16, 21, 24, 37, 43]. This category of content is commonly referred to as Deepfakes, primarily denoting audio and visual content. In this context, deepfakes are distinguished as sophisticated digitally manipulated audio and visual content portraying non-existent entities or entities in which facial or bodily features have been manipulated to simulate a different identity [2, 3, 16, 21, 33, 43, 44, 50]. The creation of deepfakes hinges upon the principles of DL [50], a subset of ML that capitalises on artificial neural networks to glean insights from substantial datasets [20, 43, 44]. Specifically, the GAN framework comprises two interconnected networks: the generator, responsible for content creation, and the discriminator, tasked with

discerning the authenticity of input images [15, 44, 50]. Research on the subject of deepfakes elucidates the realism of such content and its potential manifold effects [25, 36]. These effects consist of alterations, for instance, in user attitudes towards political figures [9] and the erosion of trust in news sources [42].

In spite of the predominant emphasis on images and videos in scholarly investigations, it is noteworthy that AI technologies, including GPT-2, hold the potential to generate textual content that closely emulates genuine news articles. Such generated content can be proliferated en masse, thereby inundating the information ecosystem with a surge of material centred around a specific subject [16, 23, 24, 44]. Empirical studies in this domain reveal that while users can often distinguish between AI and human-generated texts, AI tools still have the capability to induce a sense of confusion and cultivate mistrust effectively [16, 23, 24]. Given the rapid trajectory of technological advancement, it is becoming increasingly feasible for individuals with limited technical prowess to craft plausible disinformation [43]. The authenticity of such disinformation can be notably augmented through the amalgamation of various formats [2, 37]. For instance, [4] posited an assumption wherein:

(...) one may generate controversial text for a speech supposedly given by a political leader, create a 'deep fake' video (...) of the leader standing in the UN General Assembly delivering the speech (trained on the large amount of footage from such speeches), and then reinforce the impersonation through the mass generation of news articles allegedly reporting on the speech. (p. 4)

In summary, the rapid advancement in AI technology has raised concerns about its potential use in generating disinformation, particularly through deepfakes, which involve digitally manipulated audio and visual content. Deepfakes are created using DL, specifically the GAN framework, which consists of a content generator and an authenticity discriminator. Research on deepfakes highlights their realistic nature and their impact on user attitudes and trust in news sources. While most research focuses on image and video deepfakes, it is important to note that AI technologies such as GPT-2 can also produce highly convincing textual content resembling genuine news articles. Although users can often distinguish between AI-generated and human-generated texts, AI tools still have the capacity to sow confusion and mistrust. Finally, as technology advances, even individuals with limited technical skills can create convincing disinformation, further amplified by combining different content formats.

11.3 The Role of AI in the Dissemination of Disinformation

In addition to its potential role in generating disinformation, AI techniques can also be harnessed for the rapid dissemination of disinformation to targeted audiences [3, 9, 27, 43, 44]. Social media companies have strategically employed targeted advertising and content delivery to specific users based on their online activities, thereby increasing user engagement and platform usage, and consequently, enhancing profits derived from advertising companies [1–3, 8, 24, 27, 31, 33, 35, 43]. This practice,

however, poses certain challenges, as research indicates that users tend to engage more with content that aligns with their existing beliefs and emotional stimuli, even if the information is untrue [27, 35, 43, 45, 51]. Consequently, SMPs are likely to reinforce this behaviour by recommending more disinformation to users who have already interacted with such content [8], leading to the potential entrapment of individuals within disinformation filter bubbles [40]. These filter bubbles represent distinct information universes that surround individuals with ideas that resonate with their pre-existing beliefs, including false notions and manipulated perceptions of reality, thereby diminishing informational diversity [1, 3, 21, 31, 35, 37, 40, 45].

Furthermore, the issue is compounded by the use of social bots by malicious actors. Social bots are user accounts equipped with algorithms that autonomously post and share content on various online platforms [2, 3, 12, 14, 18, 21, 34, 35, 45]. These bots play a significant role in disseminating and amplifying disinformation by targeting vulnerable individuals and promoting trending topics [1–3, 12, 14, 16, 18, 21, 23, 33–35, 37, 45]. This phenomenon could contribute to an echo chamber effect, where individuals predominantly encounter information that aligns with their pre-existing beliefs, while opposing viewpoints are minimised. This, in turn, reinforces their pre-dispositions under the illusion that they represent the prevailing perspective [1, 14, 35, 37, 43, 45].

The impact of using technological advancements to generate and disseminate disinformation remains a subject of ongoing investigation, and the existing literature tends to converge on the notion that it exerts negative effects on both individual users and broader societies [3, 8, 16, 18, 24, 43, 50]. Notably, exposure to highly realistic disinformation, coupled with the phenomenon of filter bubbles and echo chambers, can significantly influence users' opinions, engender confusion and uncertainty, and impede their capacity to make well-informed decisions [1, 3, 21, 31, 42, 43]. The ramifications of online disinformation extend beyond individual experiences, posing threats to broader societal structures. Therefore, trust in media, news outlets, and institutional frameworks might be compromised due to the proliferation of disinformation [1–3, 12, 16, 18, 21, 24, 34, 35, 43–45, 50, 51].

In summary, AI techniques not only are capable of generating disinformation but can also rapidly disseminate it to specific audiences. Social media companies leverage targeted advertising and content delivery based on user activities, potentially reinforcing pre-existing beliefs and emotional responses, even if the information is false. This practice might trap individuals in disinformation filter bubbles, diminishing informational diversity. Additionally, malicious actors further compound the issue by using social bots to amplify and spread disinformation, potentially creating echo chambers that reinforce existing beliefs. Finally, the impact of using technology for disinformation appears to be largely negative, affecting individual opinions, causing confusion, and hindering informed decision-making. It also poses broader societal threats by eroding trust in media, news outlets, and institutional frameworks.

11.4 The Role of AI in Countering Disinformation and Associated Challenges

Due to the potential adverse effects of online disinformation on both users and democratic processes, the imperative to counter its dissemination becomes paramount. However, the sheer volume of information propagated across online platforms renders manual detection unfeasible [3, 6, 11, 21, 39, 43, 44, 50]. Consequently, the adoption of AI has emerged as an increasingly favoured approach to tackle online disinformation, owing to its capability for the automatic analysis and processing of vast datasets, thereby enhancing the efficiency and speed of disinformation detection [3, 6, 11, 13, 15, 21, 28, 39, 43, 44, 50]. Of particular note, SMPs have widely employed such AI systems to diminish the visibility and engagement with disinformation, pre-empt its dissemination, and remove it from their platforms [3, 11, 21, 27, 28, 51].

Nonetheless, as discussed in the subsequent sections, the detection and elimination of disinformation content present formidable challenges. Consequently, SMPs have prioritised the identification and removal of fake accounts, which are responsible for a significant portion of the disinformation circulating online [51]. As asserted by Meta [29], their AI tools successfully identify 99.30% of the fake accounts, leading to their removal before users report them. Despite their programming to emulate human online behavior [2, 3, 12, 14, 18, 21, 34, 35, 45], these fake accounts often exhibit discernible characteristics and behavioural patterns, such as an abnormal surge in activity during the initial dissemination of disinformation. Thus, leveraging AI techniques, SMPs have succeeded in automatically identifying these patterns and proactively removing fake accounts [1, 6, 12, 21, 33, 51].

The current literature prominently discusses the use of the same tools employed for generating fake content to also identify manipulated and autogenerated content. These tools become well-suited for detection purposes due to their familiarity with the data patterns commonly associated with autogenerated content [2, 15, 16, 23, 38, 44, 50]. Empirical investigations have substantiated the effectiveness of these AI tools in detecting indicators of manipulation, such as irregularities in eye features [7], colour components [26], noise patterns [22], and word predictability [38, 48]. Despite the ongoing evolution of these techniques, autogenerated content still exhibits artificial fingerprints, intrinsic to the generating tool, which prove challenging to erase [2, 15, 50]. Studies in this domain have demonstrated that tools designed to focus on the identification of such artifacts can effectively differentiate genuine content from autogenerated content [46, 49]. Consequently, the literature highlights the successful employment of AI tools in combating the spread of disinformation by effectively detecting autogenerated content and the bots responsible for spreading disinformation.

Despite AI's potential benefits, its use in countering online disinformation also brings limitations, challenges, and unintended consequences [5, 8, 11, 21, 39, 51]. AI technologies are still in developmental stages and currently lack the capability to

effectively detect more intricate instances of disinformation [3]. As discussed previously, disinformation is deliberately designed to appear credible, thereby deceiving, influencing, and manipulating individuals [3, 10], making it inherently challenging to differentiate between fake and genuine information. This task is further complicated within the present information ecosystem, where information conflicting with an individual's perspective is readily labelled as "fake news" [35]. Moreover, the existence of more complex forms of disinformation poses a substantial challenge for AI technologies [3]. Disinformation can be generated and disseminated in diverse formats, often adopting a multimodal nature. This, in turn, renders it more credible and poses a challenge for automatic detection, considering that most systems assume a predefined generation model [21, 27, 37, 39].

Additionally, as discussed throughout this book, AI systems might exhibit biases [5, 21, 47]. The development of these technologies involves human contributors with inherent limitations in knowledge and personal values, which might inadvertently influence the system's functioning. Moreover, the data used to train these algorithms can be incomplete, unrepresentative, or infused with human bias [3, 21, 27, 30, 47]. For instance, if an algorithm is trained on data limited to a specific context or language, it will likely fail to identify disinformation related to other topics or events or presented in another language [37]. Furthermore, AI systems could erroneously infer causality or correlation [3, 21, 27, 30, 47]. This limitation is intertwined with the complexity and opacity of AI systems. As these algorithms are self-learning in nature, they might reach a level of sophistication where even their developers cannot fully comprehend and elucidate their decision-making processes [8, 21, 47]. Consequently, if these systems make erroneous decisions, such as misclassifying and removing legitimate content, their complexity will hinder the ability to explain the rationale behind such decisions [47].

As a result, these challenges can significantly impact the efficiency and accuracy of AI systems, potentially leading to errors in their outcomes. These errors pose critical ethical concerns due to their potential adverse impacts on users and societies [21, 47]. Specifically, these issues can give rise to occurrences of false negatives and false positives [3, 8, 21, 27]. False negatives manifest as the erroneous classification of disinformation as accurate information, permitting harmful content to evade detection and circulate unchecked within online spaces [3, 27, 28, 39, 43]. Consequently, such content has the potential to engender confusion and distrust among users, indirectly influencing elections and undermining democratic processes since individuals require access to high-quality information for making informed decisions [1, 3, 21, 31, 42, 43]. Conversely, false positives pertain to the misidentification of accurate content as disinformation. Consequently, legitimate information disseminated online, such as news reports, might face removal due to misclassification. Such missteps have the potential to pose a threat to open discourse, culminating in violations of civil liberties and instances of censorship, thereby posing a challenge to democratic principles [3, 27, 28, 39, 43].

In summary, the need to counter online disinformation is crucial due to its potential adverse effects on users and democratic processes. However, the sheer volume of online content makes manual detection unfeasible, leading to the adoption of AI for

more efficient and speedy detection. AI systems are being used by SMPs to diminish disinformation's visibility and pre-empt its dissemination. Despite the benefits of AI, its use in countering disinformation also presents challenges. For instance, AI technologies might struggle with intricate forms of disinformation, and biases in the data and algorithms can influence their functioning. As a result, these challenges can lead to errors, including false positives and false negatives, with the potential to impact user trust and democratic processes.

11.5 Recommendations

This section explores various approaches and recommendations aimed at addressing the challenges posed by AI systems in the context of countering the dissemination of disinformation online. Disinformation constitutes a multifaceted and dynamic phenomenon, involving diverse actors, objectives, formats, and creation and dissemination methods [1]. Consequently, the development and implementation of comprehensive multimodal detection frameworks are necessary. These holistic multimodal detection frameworks should consist of text and visual features while analysing content, organisation, emotions, and manipulation levels [1, 2, 37]. This holistic approach could facilitate a deeper understanding of disinformation, ultimately enhancing the effectiveness and accuracy of AI models. Additionally, considering the rapid pace of innovation in disinformation production and dissemination, countermeasure tools must be consistently updated to maintain efficacy [3, 50]. In this regard, it is suggested that SMPs allow access to their repository of collected autogenerated media to utilise as training data for AI systems, ensuring the continual enhancement of detection programs [19].

Recognising that more intricate instances of disinformation are often detected through human perception, a consensus in the literature favours a combination of AI systems and human moderation [21, 28, 51]. The collaborative approach entails the automatic reporting of content or accounts by AI to human moderators, who then analyse the material based on the platforms' guidelines, rendering decisions on whether it constitutes disinformation or not [21, 51]. Moreover, the involvement of experts, particularly in cases with potential implications for civil liberties, is recommended to assess algorithm design, decision-making processes, and outcomes [28]. Therefore, to mitigate the challenge of algorithmic bias, enhanced accountability and transparency should be essential considerations [3, 8, 21, 27, 28, 35, 43]. For instance, the proposed Algorithmic Accountability Act in the United States aims to mandate companies to audit their AI systems for bias and discrimination, thereby subjecting data and model generation processes to increased scrutiny [21, 32].

Additionally, stakeholders should invest in ethical tools that facilitate the understanding and mitigation of biases within these systems [5]. Furthermore, an approach advocated in the literature involves reconfiguring the economic model of social media, which currently relies on engagement and profit [1, 11, 35, 43, 45]. As previously mentioned, disinformation thrives on SMPs due to its ability to generate high

levels of user engagement, leading recommendation algorithms to promote such content more prominently. To counteract this phenomenon, it is imperative for these platforms to implement design changes that foster interaction with diverse perspectives [1, 11, 17, 27, 35, 43, 45]. However, stakeholders must provide incentives for these changes, as alterations to platform designs can lead to reduced engagement and subsequently impact profitability.

Similarly, to effectively combat online disinformation and its propagation through AI techniques, enhanced cooperation among a diverse range of actors, including academia, SMPs, and governments on a global scale, is essential [17, 21, 27, 35, 43, 45, 51]. Collaborative efforts should involve the sharing of best practices and the development of robust tools to establish enduring and sustainable solutions [21]. When content removal measures are applied on SMPs, companies must adhere to local laws, necessitating a common understanding between governments and social media entities regarding the scope of content moderation [51]. Furthermore, platforms should engage in mutual collaboration and information-sharing regarding removed content to curtail its dissemination on other platforms [17]. It is crucial to acknowledge that disinformation represents more than just a technological predicament, thereby demanding solutions that transcend technical measures [1]. Consequently, stakeholders are advised to invest in media literacy and digital awareness initiatives to enhance critical thinking and bolster resilience against online disinformation and manipulated content among users and democratic institutions [16, 17, 21, 23, 27, 28, 43, 44].

In summary, this section explored approaches and recommendations for addressing the challenges posed by AI systems in countering online disinformation. Given the multifaceted nature of disinformation, it is essential to develop comprehensive multimodal detection frameworks that consider various content aspects. These frameworks should be regularly updated to keep pace with evolving disinformation tactics. It was also revealed that combining AI systems with human moderation was favoured to detect more complex instances of disinformation. Additionally, experts' involvement was recommended to assess algorithms and ensure transparency and accountability. Similarly, ethical tools to mitigate biases and reconfiguring the economic model of SMPs are suggested as ways to combat disinformation more effectively. In the same vein, cooperation among academia, SMPs, and governments on a global scale was considered crucial. Moreover, sharing best practices, developing robust tools, and aligning with local laws were deemed essential steps. Finally, media literacy and digital awareness initiatives were also recommended to enhance critical thinking and resilience against disinformation among users and institutions.

11.6 Discussion and Conclusion

The phenomenon of disinformation, while not novel, has evolved significantly due to the advent of novel technologies, enabling malicious actors to enhance both efficiency and scope of its dissemination [2, 3, 11, 16, 21, 23, 28, 43]. In parallel with

developments in the field of AI, the capability to automatically generate remarkably authentic disinformation is poised to become readily accessible and user-friendly [2, 3, 9, 16, 21, 24, 43]. Moreover, a considerable portion of mainstream SMPs integrates recommendation algorithms, which have been subject to critique due to their propensity to reinforce pre-existing beliefs and incentivise emotive and provocative content [8, 43]. The susceptibility of these algorithms to manipulation by malicious agents is evident, facilitated by the deployment of social bots to fabricate trending topics and direct disinformation toward vulnerable recipients, thereby augmenting its efficacy [2, 3, 12, 14, 18, 21, 34, 35, 45].

Consequently, an escalation in the dissemination of increasingly credible disinformation is anticipated within the information ecosystem. This will potentially lead individuals into the confines of filter bubbles and echo chambers, where a persistent exposure to disinformation prevails, consequently diminishing access to divergent and credible content [1, 3, 14, 21, 31, 35, 37, 40, 43, 45]. The ramifications of this phenomenon can be detrimental to both users and society at large. As a result, it often leads to confusion, eroded trust in news and institutions, and a weakening of democratic discourse. This is due to the fact that the availability of high-quality, diverse information is essential for fostering open discussions and achieving consensus [1, 3, 21, 31, 42, 43].

Given these implications, the prompt detection of disinformation and the associated disseminating accounts assumes significance in diminishing its audience and curtailing potential risks [30, 37]. However, the proliferation of online information coupled with the refinement of content manipulation technologies renders the detection and differentiation between genuine and false information increasingly complex [2].

In view of AI systems' capacity for automated analysis and processing of voluminous datasets, these technologies hold potential to enhance the efficiency and speed of online disinformation detection [3, 6, 11, 13, 15, 21, 28, 39, 43, 44, 50]. Hence, this chapter posits AI as a dual-edged instrument, capable of not only intensifying the efficacy and scope of disinformation but also augmenting the potency of countermeasures [3, 21]. However, notwithstanding the recognition of AI-powered systems as a cost-effective means to counteract online disinformation, these systems are presently in a phase of ongoing development, accompanied by an array of challenges and limitations that evoke ethical apprehensions [21, 28]. The impetus behind these measures is to mitigate the propagation of online disinformation, owing to its potential adverse impact on users and the democratic fabric [1, 3, 21, 31, 42, 43]. Nevertheless, it is noteworthy that the application of these tools might also exert repercussions upon users and societal domains [21, 28], thus potentially engendering a paradoxical outcome. Furthermore, the juxtaposition of fundamental values, including the freedom of speech and the right to access high-quality information [3], is underscored when the deployment of AI technologies is directed at addressing this issue. In addition, the complex task of striking equilibrium between these values is exacerbated by the inherent tension, as mitigating the risks affiliated with one value can inadvertently amplify those associated with the other.

In light of the negative and potentially counterproductive ramifications that the use of AI technologies for combating online disinformation could yield, the imperative emerges for the development and implementation of AI systems to be underpinned by ethical considerations [5, 16, 21, 28, 39]. The absence of such ethical grounding could potentially culminate in inadequately designed technologies that pose potential risks to users and societies [5, 16, 21, 28, 39]. Additionally, the efficacy of AI-driven measures in counteracting online disinformation and its associated impacts is contingent upon the mitigation of the challenges and limitations inherent to these approaches. As a consequence, further research is indispensable to identify the most effective strategies for mitigating the limitations intrinsic to AI technologies. Future studies should also pivot towards an in-depth analysis of the effects of autogenerated content on both users and society at large. Furthermore, given the successful application of bots in targeting disinformation to specific individuals, a potentially equivalent use case could involve the dissemination of legitimate and diverse information to users deemed more susceptible to disinformation. However, this notion warrants additional investigation that focuses on the efficacy and potential implications of such an approach.

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